Counting Sources, Not Pages: Dependence-Aware Evidence Aggregation against Illusory Consensus in Web RAG

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Abstract

Retrieval-augmented generation (RAG) over the Web treats every retrieved page as an independent witness, so a claim repeated on many pages wins. Many of those pages share one origin: syndicated articles, scraped mirrors and LLM-written content farms restate a single source in many surface forms. We call the result *illusory consensus* and show that it misleads current RAG systems. On EchoBench, a benchmark built from real web search results in which one wrong origin and its copies compete with independent correct pages, eight copies drop the accuracy of a Qwen2.5-7B reader from 86.5% to 49.5% and that of per-document majority voting to 24.5%. Near-duplicate removal and semantic deduplication recover little, because paraphrased, translated and summarised copies are not near duplicates. We propose IndieRAG, which brings copy detection from truth discovery to RAG. It estimates whether two pages share an origin from lexical, semantic, rare-fact and machine-text signals, groups pages into origin clusters, and weighs each claim by its effective number of independent sources. IndieRAG raises echo accuracy by 17.3 and 11.0 points over vanilla RAG for two 8B-class readers (10.1 and 5.8 points over the strongest of 12 baselines), keeps clean accuracy, and stays nearly flat as copies accumulate. Its shared-origin detector reaches 99.6 AUC, where deduplication reaches 54 to 60, and costs 6 ms per query on a CPU. On RAMDocs, IndieRAG improves over vanilla RAG by 6.2 and 10.4 points. On 6,397 real Bing result pages, 27.4% of the pages share an origin with another page of the same result list, 2.5 times the rate visible to near-duplicate detection.

CCS Concepts

• **Information systems** → **Question answering**; *Web mining*.

Keywords

retrieval-augmented generation, misinformation, copy detection, truth discovery, AI-generated content

1 Introduction

Truth-discovery research showed for structured data that copying between sources breaks majority voting, and that modelling the copying recovers the truth [8, 13]. RAG systems built on Web search still count pages. A reader that sees ten pages, seven of which give the same answer, takes the seven as corroboration, even when they are one article syndicated seven times. Such duplication is now cheap. News is syndicated, Q&A and trivia sites are scraped and mirrored, LLMs rewrite one page into many fluent variants at almost no cost, and neural retrievers rank LLM-written text above human-written text [6]. Corpus-poisoning attacks exploit the same weakness by injecting a handful of agreeing passages [29]. A claim that appears on many pages but comes from one origin produces *illusory consensus* (Figure 1a).

Existing defences judge passages one at a time or reward agreement between them. Work on knowledge conflicts studies how readers weigh retrieved context against parametric memory [5, 16, 19, 23, 25], and robust-RAG methods filter, isolate or re-weight passages [7, 17, 20–22, 24, 28]. Reliability estimation by cross-source agreement [10] even rewards copying, because copies always agree. Near-duplicate removal [4] and semantic deduplication [1] catch verbatim mirrors, yet the copies that LLM rewriting produces are paraphrases, translations and summaries.

We treat the problem as copy detection over anonymous web text and make four contributions.

(1) We define illusory consensus and build EchoBench from real TriviaQA search results. One wrong origin and up to eight copies of five types, written by two LLMs or by rule, compete with independent correct pages. Control settings separate repetition from LLM style and from independent disagreement.
(2) We show that four readers from two model families, up to 32B parameters, lose 16 to 20 points of accuracy on average under echoes and up to 37 points with eight copies. Majority voting over pages collapses, and deduplication and robust-RAG baselines recover part of the loss.
(3) We propose IndieRAG (Figure 1b), which estimates shared origin, clusters pages, and aggregates claims by their effective sample size (ESS), either as a vote (IndieVote), as an evidence table for the reader, or as a stopping signal for a search agent (IndieSearch). A copy-corrected Condorcet argument explains why page counting fails and how detector recall governs the repair.
(4) IndieRAG improves echo accuracy by 17.3 and 11.0 points over vanilla RAG, significantly over all 12 baselines, with unchanged clean accuracy. It also improves on RAMDocs, which has natural conflicts. On 1,000 real Bing result lists, 27.4% of the pages share an origin with another retrieved page, against 10.9% detectable as lexical near-duplicates.

2 Related Work

Truth discovery and copy detection. Dong et al. [8] showed that copying between structured sources misleads voting and estimated truth, source accuracy and copying jointly (ACCU and its copy-aware extensions); Li et al. [13] survey the area. Sharing *rare or false* values is strong evidence of copying, an observation that motivates our rare-fact signal. We move the idea from structured claims of named sources to anonymous web pages, where both claims and dependence must be inferred from free text and where LLM rewriting hides copying from string matching.

Knowledge conflicts and robust RAG. Surveys and benchmarks study conflicts between context and memory [5, 16, 19, 23, 25].

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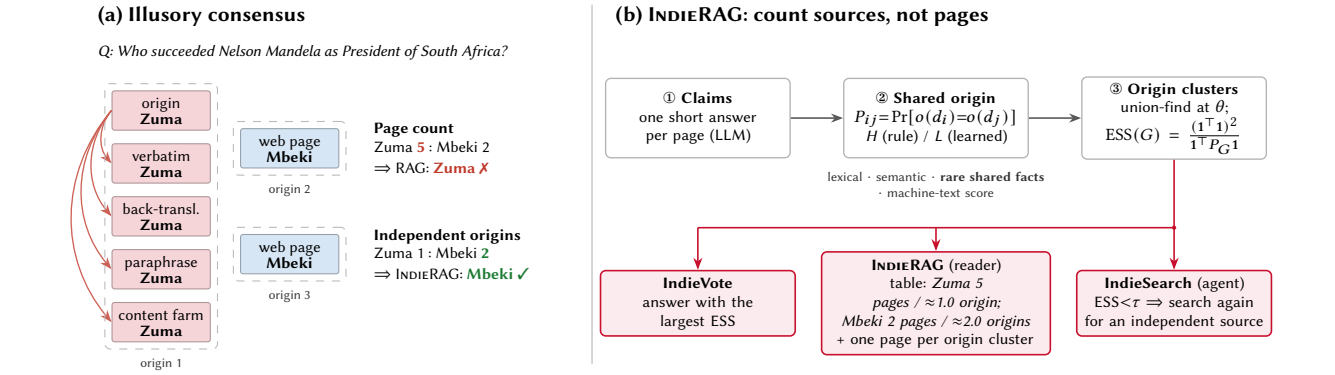


Figure 1: Overview. (a) Illusory consensus in a schematic example: one wrong origin echoed by four copies out-counts two independent correct pages. (b) IndieRAG estimates which pages share an origin and aggregates claims by their effective number of independent sources (ESS), as a vote, as an evidence table with one page per origin for the reader, or as a stopping signal for a search agent.

RAMDocs and MADAM-RAG [21] combine ambiguity, misinformation and noise and resolve them by multi-agent debate. Astute RAG [20] consolidates internal and external knowledge, InstructRAG [22] generates rationales, TrustRAG [28] filters suspicious clusters, RobustRAG [24] isolates passages and aggregates their answers, and CAG and CrAM [7, 17] condition generation on credibility. RA-RAG [10] estimates source reliability by cross-checking sources. Self-RAG and CRAG [2, 26] decide when to retrieve or correct. PoisonedRAG [29] injects several adversarial passages per target question, which is illusory consensus created on purpose.

Deduplication and machine-text detection. Shingling with containment and resemblance, approximated at scale by MinHash and LSH [4], SimHash fingerprints for web crawling [15], and embedding-based SemDeDup [1] remove near-identical documents with a binary similarity decision. Paraphrased or summarised copies fall below such thresholds, and independent pages about the same fact can exceed them. Zero-shot detectors such as Fast-DetectGPT [3] and Binoculars [9] flag LLM text; we use such a score as one weak feature.

Search agents. Search-R1 [11] and R1-Searcher [18] train agents with RL to interleave search and reasoning [14]. These agents stop once retrieved evidence agrees, and echoes make agreement cheap.

3 Problem and a Copy-Corrected Condorcet View

A question q comes with retrieved pages $D = \{d_1, \ldots, d_n\}$. Each page supports at most one candidate answer $a(d_i)$ and is generated by an unobserved *origin* $o(d_i)$; pages with the same origin are verbatim, translated, paraphrased, summarised or rewritten copies of each other. A group $G_a = \{d : a(d) = a\}$ shows *illusory consensus* when $|G_a|$ is large and the number of distinct origins $|\{o(d) : d \in G_a\}|$ is small. Page-level aggregation, explicit (majority vote) or implicit (a reader swayed by repetition), estimates $|G_a|$. The quantity that carries evidence is the number of origins.

Proposition 3.1 (Copy-corrected Condorcet). *Let n origins be independently correct with probability $p > \frac{1}{2}$ in a binary decision, and let one origin be copied m times. Let A_m be the accuracy of majority voting over pages, with ties broken uniformly. Then A_0 is the Condorcet accuracy* $\Pr[\mathrm{Bin}(n, p) > \frac{n}{2}]$ *plus half the tie mass, and $A_m = p$ for every $m \geq n - 1$: the copied origin alone decides the vote. If each copy is detected and merged into its origin independently with probability r, the accuracy is $\mathbb{E}[A_U]$ with $U \sim \mathrm{Bin}(m, 1 - r)$ undetected copies. It equals A_0 for $r = 1$ and decays to p as $m(1 - r)$ grows.*

Proof sketch. The copied origin contributes $m+1$ identical votes and the other $n-1$ origins at most $n-1$ votes. For $m+1 > n-1$ the copied origin is always in the majority, so the vote is correct exactly when that origin is, with probability p. Merging detected copies leaves U extra votes, and conditioning on U gives the mixture. □

A simulation with n=5, p=0.65 and $2 \cdot 10^5$ trials confirms this (Figure 2a). Page voting falls from 76.3% to 64.9 to 65.1% once $m \geq 4$, while a perfect detector stays at 76.3 to 76.6% for all $m \leq 32$. With recall 0.8 the accuracy at m=32 is 65.4%, so a detector has to catch nearly every copy, paraphrases included, to help against heavy echoing.

4 EchoBench

Source data. We sample 1,000 of the 6,376 TriviaQA rc.web [12] validation questions whose cached Bing results contain at least four pages from distinct domains that mention the gold answer. Up to three of these real pages, truncated to 220 words, serve as *independent correct* evidence.

Wrong origin and copies. A builder LLM (Qwen2.5-7B-Instruct) proposes a plausible wrong answer (valid for 984 questions) and rewrites another real page of the question so that it states this

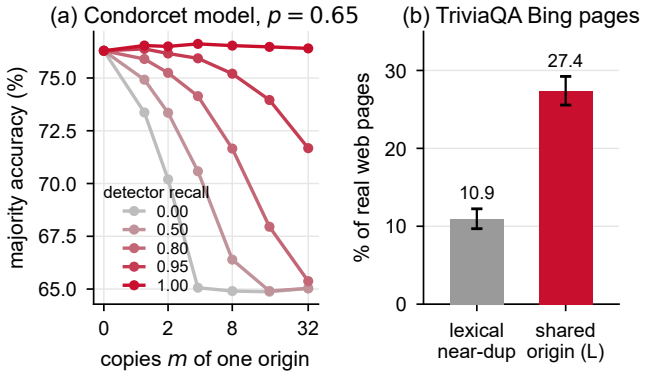


Figure 2: (a) Simulation of Proposition 3.1: majority accuracy over pages vs. copies of one origin, for detectors of increasing recall. (b) Real web pages (TriviaQA Bing results, 6,397 pages) that have a lexical near-duplicate vs. that share an origin with another page of the same result list according to IndieRAG-L (95% bootstrap CIs).

answer (a valid *origin* for 760 questions). From the origin we produce eight copies of each of five types: *verbatim* syndication (rule-based, with random site boilerplate and optional truncation), *back-translation* through one of five pivot languages, *paraphrase*, *content-farm* rewriting with one of four personas, and *summary*. Half of the paraphrase and content-farm copies come from a second generator, Phi-3.5-mini-instruct, which lets us test transfer across generators. A copy is kept only if it still states the wrong answer and does not mention the gold answer; the filter keeps 99.2% of the verbatim copies and 73 to 85% of the LLM-written ones, 25,142 copies in total. For each question we also write three *independent wrong* pages from different base pages without showing the origin, and sample 12 distractor pages from other questions. The 760 questions are split by question into 304 training questions, used only to train the dependence classifier and to select demonstrations, and 456 test questions.

Settings. **A (main):** one wrong origin with $m \in \{0, 1, 2, 4, 8\}$ mixed-type copies against $k \in \{1, 2, 3\}$ independent correct pages and two distractors, shuffled. For $k \geq 2$ the independent majority is correct for every m; for k=1 the origins are tied and the reader's own knowledge decides. **Types:** m=4 copies of a single type, k=2. **B:** $m \in \{1, 2, 3\}$ *independent* wrong pages without copies against k=2; a method should follow the independent majority here, and B-agree measures whether it does. **Style:** the correct pages are LLM-paraphrased as well, so that machine-written text no longer signals a wrong answer. **C:** the true page with $m \in \{2, 4\}$ copies against $w \in \{1, 2\}$ independent wrong pages. **Ranked:** pages in dense-retriever (bge-base) order. **Noisy:** $d \in \{0, 6, 10\}$ additional distractors. Real pages keep their true domains. Generated pages receive a domain from a list of fictitious sites and a simulated date, with copies dated after their origin; the metadata variant L+meta therefore serves only as a reference point.

HotpotQA-echo and RAMDocs. For search agents we build 300 HotpotQA [27] distractor-dev questions in which the answer-bearing gold paragraph is rewritten into a wrong origin with four copies, added to a local search index of 14,236 passages. RAMDocs [21] (500 questions) provides natural conflicts from ambiguity, misinformation and noise, with an imbalanced number of supporting documents and no synthetic copies.

5 Method: IndieRAG

Claims. The reader extracts one short answer per page with the instruction "answer only from this document, or NONE". Answers are grouped by normalised string equality, containment or token-F1≥0.6. All claim-based methods (majority vote, ACCU, IndieVote, IndieBayes and IndieRAG) share this step.

Shared-origin estimation. For every pair of pages we estimate $P_{ij} = \Pr[o(d_i) = o(d_j)]$ from four cheap feature groups. (i) *Lexical*: 5-gram and 3-gram Jaccard and 5-gram containment. (ii) *Semantic*: bge-base cosine and its gap to 3-gram overlap, which is large for paraphrases. (iii) *Rare shared facts*: independent sources share common facts but seldom the same rare details [8]. From each page we extract numbers, capitalised phrases and words with IDF≥ 4 (excluding question terms) and use the IDF-weighted overlap of the shared set, its maximum IDF and its size. (iv) *Machine text*: the minimum and maximum over the pair of a Fast-DetectGPT [3] curvature score (Qwen2.5-1.5B), plus the length ratio. **H** is a training-free rule, $P_{ij} = \max\big(c_5,\ \sigma(12(\rho_{ij} - 0.3)) \cdot \mathbb{1}[\cos_{ij} > 0.7]\big)$, with c_5 the 5-gram containment and ρ_{ij} the rare-fact overlap. **L** is a gradient-boosted classifier (300 trees) trained on 238,581 page pairs from the 304 training questions only. **LLM** asks the reader whether two passages derive from the same source, for candidate pairs with cosine≥0.75 or 3-gram Jaccard≥0.1.

Origin clusters and effective sample size. Pages are clustered by union-find over edges $P_{ij} \geq \theta$ (θ=0.5). For an answer group G with weights w (uniform here), the effective number of independent sources is the Kish effective sample size under correlation P:

$$\mathrm{ESS}(G) = \frac{(\sum_{i \in G} w_i)^2}{w_G^\top P_G\, w_G}, \qquad P_G = [P_{ij}]_{i,j \in G},\ P_{ii} = 1. \tag{1}$$

It equals $|G|$ for independent pages and 1 for $|G|$ exact copies, and changes smoothly with uncertain P.

Aggregators. **IndieVote** returns the answer with the largest ESS, breaking ties by page count. **IndieBayes** is a copy-aware ACCU-style update [8]: supporting pages are ordered by estimated accuracy and the vote of d_i is discounted by $\prod_{j \prec i}(1 - 0.9 P_{ij})$ over 5 EM iterations. **IndieRAG** gives the reader an *evidence table* that lists, for each candidate answer, the number of pages and the approximate number of independent origins, sorted by ESS, together with one representative page (the longest) per origin cluster. The prompt asks the reader to weigh independent origins and to use its own knowledge when the evidence is balanced, which resolves the tied case k=1 that a vote cannot.

IndieSearch. A ReAct-style agent [11] receives, after every search, an `<evidence_status>` block with the pages and ESS of each candidate answer. If it tries to answer while the leading answer has ESS $< \tau$ (τ=1.5), it is asked once to search again with a differently phrased query that excludes pages already seen and pages with cosine>0.9 to them.

Table 1: Illusory consensus on EchoBench (setting A). Accuracy (%) without copies (Clean), with m=8 copies of one wrong origin against k=2 independent correct pages, and averaged over all $m \geq 1$, $k \in \{1, 2, 3\}$ (Echo, 12 conditions × 200 test questions). Wrong↓: rate of outputting the injected wrong answer (Echo conditions). Slope: $d\,\mathrm{acc}/d\log_2(m{+}1)$ at k=2 (0 = flat). B-agr.: agreement with the independent majority in control setting B (m=1 or 3 independent wrong pages vs. k=2; no copies). Tok: reader tokens per instance (Qwen2.5-7B). Best and second best exclude oracle rows and L+meta (simulated page metadata); "–": method run on a reduced condition set.

	Reader: Qwen2.5-7B-Instruct						Reader: Qwen3-8B						
Method	Clean	m=8	Echo	Wrong↓	Slope	B-agr.	Clean	m=8	Echo	Wrong↓	Slope	B-agr.	Tok
Single-pass readers													
Vanilla RAG	80.7	49.5	61.0	31.9	−0.119	55.9	85.2	59.5	68.5	26.7	−0.090	59.0	2.5k
Near-dup removal	80.8	53.5	64.0	28.5	−0.106	55.6	85.3	58.0	70.4	24.7	−0.094	59.0	2.3k
SemDeDup	80.5	59.0	66.2	25.7	−0.088	55.9	85.2	65.0	73.0	22.2	−0.069	59.0	2.2k
TrustRAG (filter)	80.5	51.0	62.2	30.1	−0.114	55.6	85.2	61.5	70.1	25.0	−0.083	58.7	2.5k
Astute RAG	76.5	61.0	67.4	13.3	−0.050	45.0	70.2	26.0	46.3	**7.9**	−0.137	38.9	2.8k
InstructRAG-ICL	**87.3**	58.0	66.8	31.7	−0.102	63.3	**87.3**	69.0	73.6	34.8	−0.067	64.6	2.7k
CAG (prompted)	86.8	57.5	68.1	29.4	−0.099	65.1	84.7	68.5	73.7	22.6	−0.064	56.6	5.3k
Per-document claims + aggregation													
Majority vote	75.7	24.5	35.0	57.2	−0.203	67.9	78.2	27.5	40.6	51.6	−0.200	**68.4**	3.0k
Near-dup + vote	75.7	29.0	42.7	49.9	−0.190	67.9	78.2	31.5	47.7	45.0	−0.189	**68.4**	3.0k
SemDeDup + vote	75.7	33.5	46.8	45.9	−0.171	67.9	78.2	36.5	51.3	41.4	−0.170	**68.4**	3.0k
ACCU (no copy det.)	75.7	24.5	35.0	57.2	−0.203	67.9	78.2	27.5	40.6	51.6	−0.200	**68.4**	3.0k
MADAM-RAG	–	54.0	–	–	–	–	–	46.5	–	–	–	–	11.9k
Ours: dependence-aware vote													
IndieVote-H	75.8	75.5	63.1	30.6	−0.032	**70.5**	78.2	79.0	67.1	25.8	−0.027	**68.4**	3.0k
IndieVote-L	75.8	76.0	63.6	30.6	−0.030	**70.5**	78.0	79.0	67.1	26.0	−0.027	67.5	3.0k
IndieVote-LLM	75.7	65.0	59.2	34.7	−0.066	68.8	78.0	71.0	63.2	29.8	−0.054	65.8	10.0k
IndieBayes-L	75.8	75.5	63.1	30.9	−0.032	**70.5**	78.0	78.5	66.9	26.1	−0.029	67.5	3.0k
Ours: dependence-aware reader													
IndieRAG-H	81.3	78.0	77.5	**12.9**	−0.028	62.7	83.7	78.0	78.8	14.0	**−0.025**	59.8	4.6k
IndieRAG-L	80.3	**80.0**	**78.3**	13.2	**−0.017**	63.3	84.7	**79.5**	**79.5**	14.3	−0.027	62.1	4.9k
IndieRAG-L+meta	81.3	84.0	79.7	11.9	−0.007	61.9	84.5	82.5	81.4	12.5	−0.018	61.4	4.7k
IndieRAG-LLM	82.8	78.0	77.5	14.0	−0.025	62.2	83.8	**79.5**	78.5	15.0	**−0.025**	58.1	12.0k
Oracle dependence (upper bound)													
IndieVote-oracle	75.7	76.5	63.6	30.4	−0.029	67.9	78.2	80.5	67.5	25.7	−0.023	68.4	3.0k
IndieRAG-oracle	82.2	85.5	81.4	9.9	−0.002	64.4	84.7	85.5	83.8	10.8	−0.010	63.3	4.7k

6 Experimental Setup

Readers. Qwen2.5-7B-Instruct (primary) and Qwen3-8B run on the full grid; Granite-3.3-8B-Instruct (second family) and Qwen2.5-32B-Instruct-AWQ (scale) run settings A and B with six methods. All use vLLM, greedy decoding and an 8k context.

Baselines. Vanilla RAG. *Deduplication before reading:* 5-gram containment≥0.5, the quantity MinHash/LSH approximates [4], and SemDeDup [1] (bge cosine≥0.95). *Claims and aggregation:* majority vote (isolate-then-aggregate as in RobustRAG [24]), the same vote after near-duplicate or SemDeDup clustering ("Near-dup + vote" and "SemDeDup + vote", which are IndieVote with a binary similarity-based dependence), and ACCU without copy detection (source-accuracy truth discovery [8], the within-query analogue of RA-RAG [10]). *Robust RAG:* Astute RAG [20], InstructRAG-ICL [22], TrustRAG's filtering stage [28], MADAM-RAG [21] with one agent per document and two debate rounds, and CAG in prompted form [17], where the reader rates each page's credibility from 1 to 10 and sees the ratings. All baselines use the same reader; variants that need fine-tuning (InstructRAG-FT, CAG and CrAM training, cross-query RA-RAG) are outside our scope.

Budget. Each reader is evaluated on the first 200 test questions, 6,537 instances for the full grid. Expensive methods run on subsets of condition groups, and every average covers only conditions that all compared methods ran on ("–" otherwise); MADAM-RAG, for example, ran at $m \in \{0, 2, 8\}$, k=2.

Metrics and statistics. *Accuracy*: the output contains a gold alias. *Wrong*: it contains the injected wrong answer. On RAMDocs, accuracy requires all valid answers and no misinformation answer. Paired differences use a question-clustered bootstrap with 5,000 resamples, since instances of one question are dependent, and Holm correction over the 12 baselines. An LLM judge (Qwen3-8B) regrading 700 EchoBench predictions agrees with string matching on 97.4% of them (Cohen κ=0.94, ranking Kendall τ=0.80). On RAMDocs agreement is 81.4% (κ=0.56), mostly because string matching

misses paraphrased multi-answers, and the method ranking is preserved (τ=0.92; Table 9).

7 Results

7.1 Echoes mislead RAG readers

Accuracy falls with every added copy. With k=2 independent correct pages, Qwen2.5-7B's accuracy falls from 86.5% at m=0 to 71.0%, 61.5% and 49.5% at m=2, 4, 8 (Figure 4b). Three independent correct pages still lose ground to one origin echoed eight times (87.5% to 52.5%, Figure 4c). Qwen3-8B follows the same curve (88.0% to 59.5% at k=2), and so does the 32B reader, which drops from 88.0% clean to 63.0% at m=8 (Table 11). The wrong outputs are the echoed answer: 31.9% of all echo-condition outputs contain it (Table 1).

Voting over pages amplifies the problem. Per-document majority vote, the literal form of page counting, collapses to 24.5% at m=8 with a slope of −0.203, against −0.119 for vanilla RAG. ACCU-style truth discovery without copy detection is identical to it in every condition, because copies agree with each other and so appear reliable.

Deduplication removes only verbatim echoes. Near-duplicate removal raises accuracy with verbatim copies from 70.9% to 81.9% and leaves paraphrases (57.6%) and content-farm rewrites (63.6%) at the vanilla level (Figure 3b). SemDeDup also catches back-translations (82.3%) and misses content-farm copies (66.0%) and summaries (66.7%). Over all echo conditions, the two raise accuracy from 61.0% to 64.0% and 66.2%.

Robust-RAG baselines help partially and depend on the reader. Astute RAG answers conservatively, with the lowest wrong-answer rate among baselines (13.3%), and reaches 67.4% echo accuracy with Qwen2.5-7B, 46.3% with Qwen3-8B and 27.1% with Granite. InstructRAG-ICL and prompted CAG have the best clean accuracy (87.3% and 86.8%) and still slope down (−0.102 and −0.099). MADAM-RAG reaches 54.0% at m=8 with 4.8 times the tokens of vanilla RAG.

7.2 Counting sources restores accuracy

Main result. IndieRAG-L reaches 78.3% echo accuracy with Qwen2.5-7B and 79.5% with Qwen3-8B, against 61.0% and 68.5% for vanilla RAG and 68.1% and 73.7% for the strongest baseline, prompted CAG. The paired gains are +17.3 pp [13.4, 21.0] and +11.0 pp [7.5, 14.7] over vanilla RAG and +10.1 pp [6.5, 13.7] and +5.8 pp [2.5, 9.1] over CAG, and all 24 comparisons with the 12 baselines are significant after Holm correction (Table 2). The slope flattens from −0.119 to −0.017, the wrong-answer rate falls from 31.9% to 13.2%, and clean accuracy is unchanged (80.3% vs. 80.7%; 84.7% vs. 85.2%). The training-free estimator H reaches 77.5% and 78.8%, so most of the gain is available without training data, and oracle origins reach 81.4% and 83.8%.

The vote and the reader complement each other. IndieVote-L turns the collapsing majority vote into a flat curve (76.0% at m=8, k=2, against 24.5%) and follows the independent majority in setting B more often than majority vote does (B-agree 70.5% vs. 67.9%), so genuine agreement between independent pages still counts. With k=1 the wrong origin and the correct page are one origin each, and the vote falls back to page counts (24.5% at m=8), which lowers its echo average to 63.6%. IndieRAG hands this tie to the reader's knowledge and keeps 67.0% at k=1, m=8, against 36.0% for vanilla RAG.

Table 2: Paired gains of IndieRAG-L over each baseline (Δacc in points). EchoBench: setting A, $m \geq 1$ (2,400 instances; 400 for MADAM-RAG); RAMDocs: 500 questions. Question-clustered bootstrap, Holm-corrected over the 12 baselines: $^{\ddagger}p_{\text{Holm}}<0.01$, $^{\dagger}p_{\text{Holm}}<0.05$; green/red = significant gain/loss, gray = not significant. CIs in Appendix A.

	EchoBench		RAMDocs	
vs. baseline	Qwen2.5	Qwen3	Qwen2.5	Qwen3
Vanilla RAG	+17.3 ‡	+11.0 ‡	+6.2 †	+10.4 ‡
Near-dup removal	+14.3 ‡	+9.1 ‡	−3.4	−1.4
SemDeDup	+12.0 ‡	+6.5 ‡	+2.8	+6.6 †
Majority vote	+43.3 ‡	+38.9 ‡	+0.6	+10.0 ‡
Near-dup + vote	+35.6 ‡	+31.8 ‡	−0.2	+9.2 ‡
SemDeDup + vote	+31.5 ‡	+28.2 ‡	+0.6	+10.0 ‡
Astute RAG	+10.8 ‡	+33.2 ‡	+4.6	+28.6 ‡
InstructRAG-ICL	+11.4 ‡	+5.9 ‡	+5.2	+4.0
TrustRAG (filter)	+16.1 ‡	+9.4 ‡	+5.8 †	+10.8 ‡
MADAM-RAG	+17.8 ‡	+26.8 ‡	+6.8 †	+20.2 ‡
CAG (prompted)	+10.1 ‡	+5.8 ‡	+7.8 †	+7.0 ‡
ACCU (no copy det.)	+43.3 ‡	+38.9 ‡	+6.8 †	+16.2 ‡

Table 3: Controls and stress tests (Qwen2.5-7B, accuracy % unless noted). *Style*: correct pages also LLM-rewritten (m=8). *Rank.*: documents in dense-retriever order (m=8). *Noise*: 10 distractor pages (m=4). *B-agr.*: follows the independent majority when wrong sources are independent. *Setting C*: copies of the *true* page vs. w independent wrong pages; C-ind.: agreement with the independent majority at m=4, w=2. Questions per condition: style 183, ranked and noise 200, setting C 93, 98 and 68. Shaded: ours.

	Echoed misinformation				Setting C (echoed truth)		
Method	Style	Rank.	Noise	B-agr.	$m4, w1$	$m2, w2$	C-ind.
Vanilla RAG	47.5	56.5	57.0	55.9	87.1	78.6	10.3
SemDeDup	56.3	–	65.5	55.9	–	–	–
Majority vote	21.9	24.0	26.0	67.9	**93.5**	**85.7**	5.9
SemDeDup + vote	31.1	33.5	40.5	67.9	**93.5**	79.6	5.9
Astute RAG	63.4	54.5	–	45.0	84.9	**85.7**	5.9
CAG (prompted)	61.2	–	–	65.1	92.5	84.7	10.3
IndieVote-L	74.9	76.5	76.5	**70.5**	**93.5**	44.9	**45.6**
IndieBayes-L	73.8	76.5	75.5	**70.5**	**93.5**	44.9	44.1
IndieRAG-L	**77.0**	**79.0**	**81.0**	63.3	75.3	64.3	27.9

Controls for style, order and noise. All wrong pages in EchoBench are LLM-written, so a method could learn that machine-written text is wrong. When the correct pages are LLM-rewritten as well, IndieRAG-L keeps 77.0% at m=8 against 47.5% for vanilla RAG (Table 3, Figure 4d), and dropping the machine-text feature changes IndieVote-L's echo accuracy by 0.0 and +0.1 points (Table 4). Retriever-ordered pages (79.0% vs. 56.5% at m=8) and ten extra distractors (81.0% vs. 57.0%) give the same picture. Setting C shows

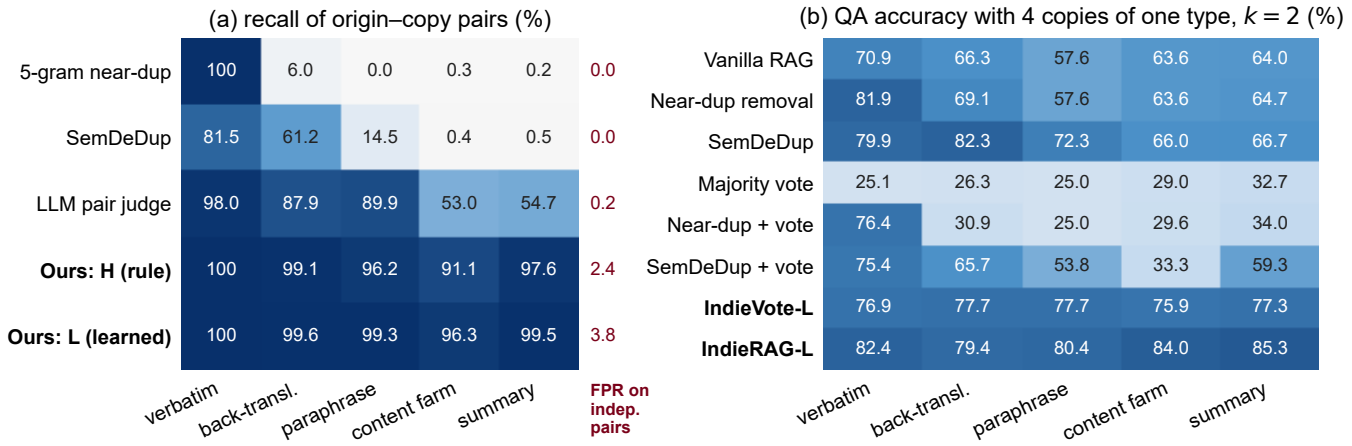


Figure 3: By copy type. (a) Recall of origin–copy pairs at threshold 0.5 per estimator, and false-positive rate on independent pairs (right column). (b) Accuracy with four copies of a single type (k=2, Qwen2.5-7B; 150 to 199 questions per type). Lexical and embedding deduplication miss paraphrase, content-farm and summary copies, which the learned estimator finds.

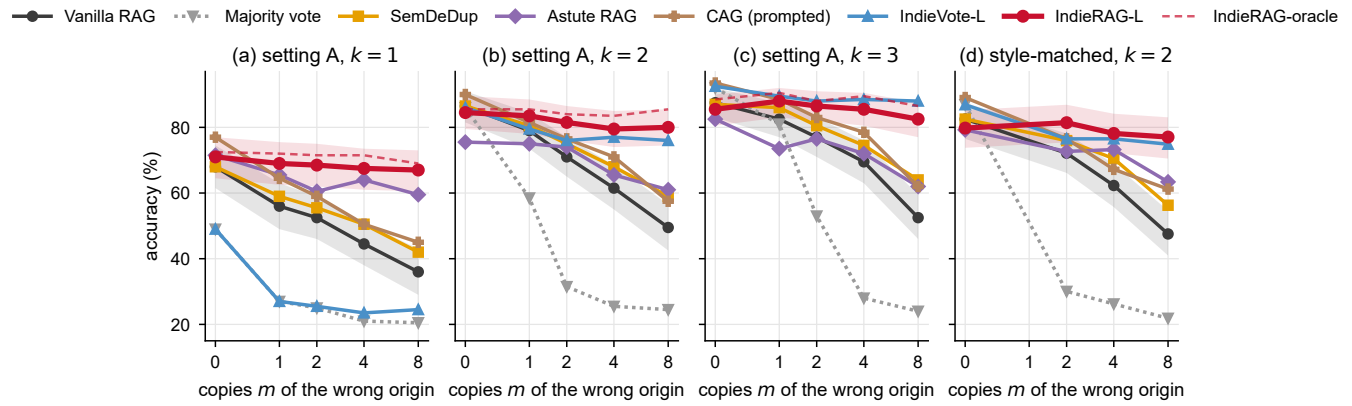


Figure 4: Accuracy vs. number of copies of one wrong origin (Qwen2.5-7B). (a to c) Setting A with k independent correct pages, 200 test questions per point; (d) style-matched control in which the correct pages are LLM-rewritten as well, 183 questions per point. Shaded: 95% bootstrap bands for vanilla RAG and IndieRAG-L. Majority vote (dotted) collapses; IndieRAG-L (red) stays nearly flat and close to the oracle-dependence upper bound (dashed).

what the method optimises. When the true page is the echoed one and two independent pages disagree with it, the independent majority is wrong; IndieVote follows the independent majority as designed (44.9% accuracy at m=2, w=2), page counting follows the echo (85.7%), and IndieRAG-L, whose reader can override a balanced table, lies in between (64.3%).

Grouping pages by origin is the main mechanism. Table 4 varies the reader input. With the evidence table but all pages in context, echo accuracy is 62.5%: the reader is still swayed by the copies it reads. Showing one page per origin cluster gives 79.2% even without the table. The table adds 1.5 points of clean accuracy and helps on RAMDocs (32.2% vs. 27.8% and 41.0% vs. 36.2%, Table 14). Among the features, the rare-fact group matters most end to end (63.6% to 62.9% and 67.1% to 66.3% without it), and a lexical-only classifier loses 2.0 and 1.3 points. The clustering threshold matters little: at m=8, k=2, $\theta \in \{0.3, 0.5, 0.7\}$ gives 77.5, 80.0 and 81.5%.

7.3 Across readers and on natural conflicts

Figure 5 summarises the four readers. At 32B, IndieRAG-L reaches 87.3% echo accuracy against 72.1% for vanilla RAG and 84.4% for Astute RAG, the strongest baseline there. The vote and the reader variant suit different readers: Granite-3.3-8B makes poor use of the grouped prompt (63.2% clean accuracy with it, 85.8% without), and for this reader the prompt-free IndieVote-L is the better choice (68.6% vs. 67.0% for vanilla RAG). On RAMDocs, with natural conflicts and no synthetic copies, IndieRAG-L improves over vanilla RAG by +6.2 pp (Holm p=0.020) and +10.4 pp (p=0.002) and over MADAM-RAG, the method proposed with RAMDocs, by +6.8 and +20.2 points; 5-gram deduplication performs on par here (35.6% and 42.4%, differences n.s.; Table 14), which suggests that RAMDocs' imbalanced support comes mostly from near-identical documents.

Table 4: Ablations (accuracy %). Echo columns: red/green = at least 1 point below/above the first row of the block. ms: dependence estimation per instance (CPU). *Page metadata in EchoBench is simulated (see §4), so L+meta is optimistic.

	Qwen2.5-7B			Qwen3		
Variant	Clean	Echo	B-agr.	Echo	ms	Tok
Reader input (IndieRAG-L)						
IndieRAG-L (full)	**80.3**	**78.3**	**63.3**	**79.5**	**6**	**4.9k**
table + all docs	82.3	**62.5**	61.5	**67.3**	6	5.3k
grouped docs, no table	78.8	79.2	62.1	80.4	6	4.6k
Dependence estimator (IndieRAG)						
H (rule)	**81.3**	**77.5**	**62.7**	**78.8**	**3**	**4.6k**
LLM pair judge	82.8	77.5	62.2	78.5	121	12.0k
L + page metadata*	81.3	**79.7**	61.9	**81.4**	5	4.7k
oracle origins	82.2	**81.4**	64.4	**83.8**	0	4.7k
Feature groups (IndieVote-L)						
IndieVote-L (all features)	**75.8**	**63.6**	**70.5**	**67.1**	**6**	**3.0k**
−lexical	75.8	63.5	70.5	67.2	6	3.0k
−semantic	75.8	63.4	70.5	67.2	6	3.0k
−rare facts	75.8	62.9	69.6	66.3	6	3.0k
−machine-text score	75.8	63.6	69.6	67.2	6	3.0k
lexical only	75.7	**61.6**	69.6	**65.8**	6	3.0k

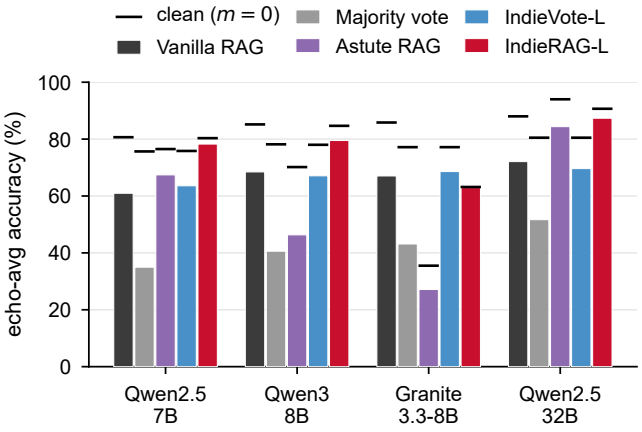


Figure 5: Echo accuracy (bars) and clean accuracy (black ticks) across reader families and scales. IndieVote-L needs no reader prompt and beats majority vote for every reader; IndieRAG-L gives the largest gains for readers that use its evidence table.

7.4 Shared origin is detectable

On the 456 test pools, L reaches 99.6 AUC (95% CI [99.5, 99.7]) and an origin-clustering ARI of 0.882 (Table 5). The deduplication estimators reach 53.8 and 60.0 AUC: they find verbatim copies (100% and 81.5% recall) and almost none of the content-farm copies or summaries (at most 0.5%, Figure 3a). The LLM pair judge is precise (0.2% false positives), recalls 53.0% of content-farm and 54.7% of summary copies, and takes 5.4 s per pool against 24 ms for L on a CPU. Rare shared facts carry the most information (ARI 0.882 to 0.799 without them), and 25 training questions already give 99.5 AUC. The hardest negatives are independent pages that state the *same* wrong answer as the copies (8.9% false positives) or the same correct answer (4.9%).

Table 5: Shared-origin detection on the pools of the 456 EchoBench test questions (189,746 pairs, 59.5% dependent). AUC/AP/F1@0.5 in %; ARI of the origin clustering at θ=0.5; ms per pool. *Simulated metadata.

Estimator	AUC	AP	F1	ARI	ms
Estimators					
5-gram near-dup (≥0.5)	53.8	62.5	14.2	0.096	1
SemDeDup (cos ≥0.95)	60.0	67.6	33.4	0.320	0
LLM pair judge (Qwen2.5-7B)	95.0	94.5	76.2	**0.923**	5388
Ours: H (rule, no training)	99.2	99.4	96.9	0.904	26
Ours: L (learned)	**99.6**	**99.7**	**97.7**	0.882	24
Ours: L + metadata*	99.9	99.9	99.0	0.949	24
Ablations and learning curve of L					
L −lexical	99.6	99.7	97.7	0.880	24
L −semantic	99.6	99.7	97.6	0.872	24
L −rare facts	99.1	99.3	96.4	0.799	24
L −machine-text score	99.6	99.7	97.8	0.881	24
L lexical only	97.9	98.5	94.4	0.763	24
L, 25 training questions	99.5	99.6	96.9	0.907	24
L, 100 training questions	99.5	99.6	97.4	0.882	24

Transfer. Trained without any Phi-3.5 copies, L still recalls 94.2% of that generator's content-farm copies and 93.3% of its paraphrases (98.2% and 97.9% when trained on them). Applied unchanged to HotpotQA-echo, whose Wikipedia paragraphs differ in style from web pages, L keeps 97.8 AUC with an AP of 73.6. The ESS computed from L is off by 0.12 origins on average, against 3.0 for page counts. In our pools the dense retriever ranks copies and correct pages similarly (mean normalised ranks 0.42 to 0.55 vs. 0.425), and because copies are numerous, 85.5% of top-3 lists contain at least one LLM-written copy.

7.5 Illusory consensus on real search results

We apply L, trained only on EchoBench, to the unmodified cached Bing results of 1,000 further TriviaQA questions (6,397 pages, 6.4 per question). 27.4% [25.6, 29.2] of the pages fall into a dependent cluster, against 10.9% [9.7, 12.2] with a lexical near-duplicate (Figure 2b), and the gold answer is supported by 1.45 times as many pages as independent origins. These are model estimates, since real pages carry no origin labels. In this 2017 snapshot the dependence seldom changes the winning answer: of the 297 questions whose pages support two or more answers, page and origin majorities disagree in 3.7%.

On these pages IndieVote-L matches vanilla RAG (91.1% vs. 91.3% and 91.3% vs. 92.1%, both n.s.; Table 6). IndieRAG-L trades 2.1 and 2.4 points on all questions for its robustness to echoes, and on the conflict questions its difference to vanilla RAG is not significant (−1.3 and +1.4 pp). We therefore recommend the reader variant where echoed misinformation is expected, for example in poisoning-prone corpora, and IndieVote-L where echoes are rare.

Table 6: Natural data: unmodified web search results (TriviaQA, cached Bing pages; 1,000 questions, 297 of them with pages supporting ≥2 answers = "conf."). Accuracy (%) on all questions and paired difference to vanilla RAG (Δ, points) on all and on conflict questions (vanilla: 84.2/84.3%). Green/red: 95% CI above/below 0; gray: CI contains 0.

	Qwen2.5-7B			Qwen3-8B		
Method	Acc	Δ	Δconf.	Acc	Δ	Δconf.
Vanilla RAG	91.3	–	–	92.1	–	–
Near-dup removal	90.8	**−0.5**	**−1.3**	91.8	−0.3	−0.4
SemDeDup	91.1	−0.2	−0.7	91.9	−0.2	+0.0
Majority vote	91.0	−0.3	−2.7	91.4	−0.7	−0.4
SemDeDup + vote	90.9	−0.4	−3.0	91.5	−0.6	+0.0
Astute RAG	80.0	**−11.3**	**−11.4**	68.9	**−23.2**	**−22.8**
InstructRAG-ICL	89.9	−1.4	+0.3	92.8	+0.7	**+5.0**
TrustRAG (filter)	90.9	**−0.4**	−0.7	91.9	−0.2	+0.0
CAG (prompted)	91.9	+0.6	+2.0	91.4	−0.7	+1.4
IndieVote-L	91.1	−0.2	−2.4	91.3	−0.8	−0.7
IndieBayes-L	91.1	−0.2	−2.4	91.4	−0.7	−0.4
IndieRAG-L	89.2	**−2.1**	−1.3	89.7	**−2.4**	+1.4

7.6 Search agents and cost

With four copies in the index, IndieSearch answers 77.7% of EchoBench questions correctly against 73.3% for ReAct with the same 7B policy and 67.7% for the RL-trained Search-R1, and gives the echoed answer less often (14.7% vs. 16.7% and 23.0%; Table 7). The paired gain over ReAct is +4.3 pp [−0.3, +9.3] (p=0.080, 300 questions). The seek-again rule carries the gain: evidence-status feedback alone (τ=0) reaches 69.0%. On multi-hop HotpotQA-echo, prompted agents perform alike and Search-R1, trained on HotpotQA, is best. IndieRAG-L costs 9.5 LLM calls per instance (8.5 claim extractions with 17 generated tokens in total, plus one reader call), 4.9k prompt tokens and two sequential rounds; dependence estimation takes 6 ms on a CPU (Table 12, Figure 7). This is twice the tokens of vanilla RAG, similar to prompted CAG (5.2k), and less than half of MADAM-RAG (11.2k tokens, 18.9 calls).

8 Scope and Ethics

EchoBench copies come from two 4B to 7B LLMs and from rules, and copy validity is checked automatically (wrong answer kept, gold answer absent); larger content-farm models may rewrite more freely, which the cross-generator test and the natural-data results address in part. IndieRAG measures independent support, so it follows independent majorities and discounts echoed truths by design (settings B and C). The RAG readers were evaluated on 200 test questions each, and the natural-data study uses a reproducible 2017 snapshot that predates LLM content farms. Shared-origin signals could help adversaries diversify poisoned copies; the same signals make cheap mass rewriting less effective. We will release the benchmark and code for defensive research. All source pages come from the public TriviaQA release.

9 Conclusion

RAG readers, robust-RAG methods and deduplication all treat echoes of one origin as corroboration. IndieRAG estimates shared origin from text and aggregates claims by their effective number of independent sources. On EchoBench this keeps accuracy nearly flat as copies accumulate, with clean accuracy unchanged, and the detector finds the paraphrased and content-farm copies that deduplication misses in milliseconds per query. On real Bing results it places more than a quarter of the pages in an origin cluster with another retrieved page.

References

[1] Amro Abbas, Kushal Tirumala, Dániel Simig, Surya Ganguli, and Ari S. Morcos. 2023. SemDeDup: Data-efficient Learning at Web-scale through Semantic Deduplication. *arXiv preprint arXiv:2303.09540* (2023).

[2] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. *arXiv preprint arXiv:2310.11511* (2023).

[3] Guangsheng Bao, Yanbin Zhao, Zhiyang Teng, Linyi Yang, and Yue Zhang. 2024. Fast-DetectGPT: Efficient Zero-Shot Detection of Machine-Generated Text via Conditional Probability Curvature. In *International Conference on Learning Representations (ICLR)*.

[4] Andrei Z. Broder. 1997. On the Resemblance and Containment of Documents. In *Proceedings of Compression and Complexity of SEQUENCES 1997*. 21–29. doi:10.1109/SEQUEN.1997.666900

[5] Yihang Chen, Pin Qian, Su Wang, Sipeng Zhang, Huan Xu, Shuhuai Lin, and Xinpeng Wei. 2026. Does RAG Know When Retrieval Is Wrong? Diagnosing Context Compliance under Knowledge Conflict. *arXiv preprint arXiv:2605.14473* (2026).

[6] Sunhao Dai, Yuqi Zhou, Liang Pang, Weihao Liu, Xiaolin Hu, Yong Liu, Xiao Zhang, Gang Wang, and Jun Xu. 2024. Neural Retrievers are Biased Towards LLM-Generated Content. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. arXiv:2310.20501.

[7] Boyi Deng, Wenjie Wang, Fengbin Zhu, Qifan Wang, and Fuli Feng. 2024. CrAM: Credibility-Aware Attention Modification in LLMs for Combating Misinformation in RAG. *arXiv preprint arXiv:2406.11497* (2024).

[8] Xin Luna Dong, Laure Berti-Équille, and Divesh Srivastava. 2009. Integrating Conflicting Data: The Role of Source Dependence. *Proceedings of the VLDB Endowment* 2, 1 (2009), 550–561. doi:10.14778/1687627.1687690

[9] Abhimanyu Hans, Avi Schwarzschild, Valeriia Cherepanova, Hamid Kazemi, Aniruddha Saha, Micah Goldblum, Jonas Geiping, and Tom Goldstein. 2024. Spotting LLMs With Binoculars: Zero-Shot Detection of Machine-Generated Text. In *International Conference on Machine Learning (ICML)*.

[10] Jeongyeon Hwang, Junyoung Park, Hyejin Park, Dongwoo Kim, Sangdon Park, and Jungseul Ok. 2024. Retrieval-Augmented Generation with Estimation of Source Reliability. *arXiv preprint arXiv:2410.22954* (2024).

[11] Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Sercan Ö. Arık, Dong Wang, Hamed Zamani, and Jiawei Han. 2025. Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning. In *Conference on Language Modeling (COLM)*.

[12] Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. 2017. TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*.

[13] Yaliang Li, Jing Gao, Chuishi Meng, Qi Li, Lu Su, Bo Zhao, Wei Fan, and Jiawei Han. 2015. A Survey on Truth Discovery. *arXiv preprint arXiv:1505.02463* (2015).

[14] Minhua Lin, Zongyu Wu, Zhichao Xu, Hui Liu, Xianfeng Tang, Qi He, Charu Aggarwal, Hui Liu, Xiang Zhang, and Suhang Wang. 2025. A Comprehensive Survey on Reinforcement Learning-based Agentic Search: Foundations, Roles, Optimizations, Evaluations, and Applications. *arXiv preprint arXiv:2510.16724* (2025).

[15] Gurmeet Singh Manku, Arvind Jain, and Anish Das Sarma. 2007. Detecting Near-Duplicates for Web Crawling. In *Proceedings of the 16th International Conference on World Wide Web (WWW)*. 141–150. doi:10.1145/1242572.1242592

[16] Yifei Ming, Senthil Purushwalkam, Shrey Pandit, Zixuan Ke, Xuan-Phi Nguyen, Caiming Xiong, and Shafiq Joty. 2025. FaithEval: Can Your Language Model Stay Faithful to Context, Even If "The Moon is Made of Marshmallows". In *International Conference on Learning Representations (ICLR)*.

[17] Ruotong Pan, Boxi Cao, Hongyu Lin, Xianpei Han, Jia Zheng, Sirui Wang, Xunliang Cai, and Le Sun. 2024. Not All Contexts Are Equal: Teaching LLMs Credibility-aware Generation. *arXiv preprint arXiv:2404.06809* (2024).

[18] Huatong Song, Jinhao Jiang, Yingqian Min, Jie Chen, Zhipeng Chen, Wayne Xin Zhao, Lei Fang, and Ji-Rong Wen. 2025. R1-Searcher: Incentivizing the Search

Table 7: Search agents (300 questions each, local dense search engine). m: copies of a wrong origin injected into the index. #S: searches per question.

	EchoBench				HotpotQA-echo		
Agent	m=0	m=4	Wr.↓	#S	m=0	m=4	#S
ReAct (Qwen2.5-7B)	84.0	73.3	16.7	0.77	40.3	25.3	1.33
Search-R1 (RL, 7B)	**87.3**	67.7	23.0	1.57	**47.0**	**32.0**	2.42
IndieSearch (τ=1.5)	**87.3**	**77.7**	**14.7**	2.04	41.3	26.7	3.29
feedback only (τ=0)	86.3	69.0	20.7	1.30	41.3	24.0	2.37
stricter (τ=2)	86.3	73.7	18.3	2.55	41.3	27.0	3.34

Capability in LLMs via Reinforcement Learning. *arXiv preprint arXiv:2503.05592* (2025).

[19] Zhaochen Su, Jun Zhang, Xiaoye Qu, Tong Zhu, Yanshu Li, Jiashuo Sun, Juntao Li, Min Zhang, and Yu Cheng. 2024. ConflictBank: A Benchmark for Evaluating the Influence of Knowledge Conflicts in LLM. *arXiv preprint arXiv:2408.12076* (2024).

[20] Fei Wang, Xingchen Wan, Ruoxi Sun, Jiefeng Chen, and Sercan Ö. Arık. 2024. Astute RAG: Overcoming Imperfect Retrieval Augmentation and Knowledge Conflicts for Large Language Models. *arXiv preprint arXiv:2410.07176* (2024).

[21] Han Wang, Archiki Prasad, Elias Stengel-Eskin, and Mohit Bansal. 2025. Retrieval-Augmented Generation with Conflicting Evidence. In *Conference on Language Modeling (COLM).*

[22] Zhepei Wei, Wei-Lin Chen, and Yu Meng. 2025. InstructRAG: Instructing Retrieval-Augmented Generation via Self-Synthesized Rationales. In *International Conference on Learning Representations (ICLR).*

[23] Kevin Wu, Eric Wu, and James Zou. 2024. ClashEval: Quantifying the tug-of-war between an LLM's internal prior and external evidence. *arXiv preprint arXiv:2404.10198* (2024).

[24] Chong Xiang, Tong Wu, Zexuan Zhong, David Wagner, Danqi Chen, and Prateek Mittal. 2024. Certifiably Robust RAG against Retrieval Corruption. *arXiv preprint arXiv:2405.15556* (2024).

[25] Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. 2024. Knowledge Conflicts for LLMs: A Survey. *arXiv preprint arXiv:2403.08319* (2024).

[26] Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. 2024. Corrective Retrieval Augmented Generation. *arXiv preprint arXiv:2401.15884* (2024).

[27] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP).*

[28] Huichi Zhou, Kin-Hei Lee, Zhonghao Zhan, Zhenhao Li, Yue Chen, Zhaoyang Wang, Hamed Haddadi, and Emine Yilmaz. 2025. TrustRAG: Enhancing Robustness and Trustworthiness in Retrieval-Augmented Generation. *arXiv preprint arXiv:2501.00879* (2025).

[29] Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia. 2025. PoisonedRAG: Knowledge Corruption Attacks to Retrieval-Augmented Generation of Large Language Models. In *Proceedings of the 34th USENIX Security Symposium.*

A Additional Results

This appendix gives the search-agent results, the full significance tests with confidence intervals, the LLM-judge validation, results per copy type and per reader, the cost breakdown, the controls for the second reader, RAMDocs, the natural-data comparison with confidence intervals, and the accuracy-cost plot.

B Reproducibility

One script runs the complete pipeline (benchmark construction, all methods and statistics) as resumable jobs and caches every LLM call. The tables in this paper are generated from the result files by a script that copies numbers verbatim, and the figures are drawn from the raw results and per-instance predictions. Hyper-parameters: 220-word passages; 8 copies per type; θ=0.5; τ=1.5; SemDeDup cosine 0.95; near-duplicate 5-gram containment 0.5; agents use the top-3 results and at most 4 searches; bootstrap with 5,000 resamples clustered by question; seed 2026.

Table 8: Paired significance of IndieRAG-L vs. every baseline: Δacc, 95% question-clustered bootstrap CI, bootstrap p, Holm-corrected p, and McNemar p (reference).

vs. baseline	n	Δacc	95% CI	p	p_{Holm}	p_{McN}
EchoBench, Qwen2.5-7B						
Vanilla RAG	2400	+17.3	[+13.4, +21.0]	<0.001	0.002	<0.001
Near-dup removal	2400	+14.3	[+10.6, +18.0]	<0.001	0.002	<0.001
SemDeDup	2400	+12.0	[+8.5, +15.7]	<0.001	0.002	<0.001
Majority vote	2400	+43.3	[+37.8, +48.6]	<0.001	0.002	<0.001
Near-dup + vote	2400	+35.6	[+30.5, +40.7]	<0.001	0.002	<0.001
SemDeDup + vote	2400	+31.5	[+26.6, +36.3]	<0.001	0.002	<0.001
Astute RAG	2400	+10.8	[+6.0, +15.4]	<0.001	0.002	<0.001
InstructRAG-ICL	2400	+11.4	[+8.0, +15.0]	<0.001	0.002	<0.001
TrustRAG (filter)	2400	+16.1	[+12.3, +19.8]	<0.001	0.002	<0.001
MADAM-RAG	400	+17.8	[+11.5, +24.0]	<0.001	0.002	<0.001
CAG (prompted)	2400	+10.1	[+6.5, +13.7]	<0.001	0.002	<0.001
ACCU (no copy det.)	2400	+43.3	[+37.8, +48.6]	<0.001	0.002	<0.001
EchoBench, Qwen3-8B						
Vanilla RAG	2400	+11.0	[+7.5, +14.7]	<0.001	0.002	<0.001
Near-dup removal	2400	+9.1	[+5.8, +12.6]	<0.001	0.002	<0.001
SemDeDup	2400	+6.5	[+3.2, +9.7]	<0.001	0.002	<0.001
Majority vote	2400	+38.9	[+33.8, +44.0]	<0.001	0.002	<0.001
Near-dup + vote	2400	+31.8	[+26.8, +36.9]	<0.001	0.002	<0.001
SemDeDup + vote	2400	+28.2	[+23.5, +33.0]	<0.001	0.002	<0.001
Astute RAG	2400	+33.2	[+28.8, +37.4]	<0.001	0.002	<0.001
InstructRAG-ICL	2400	+5.9	[+2.5, +9.4]	<0.001	0.002	<0.001
TrustRAG (filter)	2400	+9.4	[+5.8, +13.0]	<0.001	0.002	<0.001
MADAM-RAG	400	+26.8	[+20.0, +33.8]	<0.001	0.002	<0.001
CAG (prompted)	2400	+5.8	[+2.5, +9.1]	0.001	0.002	<0.001
ACCU (no copy det.)	2400	+38.9	[+33.8, +44.0]	<0.001	0.002	<0.001
RAMDocs, Qwen2.5-7B						
Vanilla RAG	500	+6.2	[+2.4, +10.0]	0.002	0.020	0.002
Near-dup removal	500	−3.4	[−7.8, +0.8]	0.123	0.614	0.135
SemDeDup	500	+2.8	[−1.2, +6.8]	0.184	0.736	0.206
Majority vote	500	+0.6	[−3.8, +4.8]	0.816	1.000	0.856
Near-dup + vote	500	−0.2	[−4.6, +3.8]	0.964	1.000	1.000
SemDeDup + vote	500	+0.6	[−3.8, +4.8]	0.816	1.000	0.856
Astute RAG	500	+4.6	[+0.6, +8.6]	0.034	0.204	0.042
InstructRAG-ICL	500	+5.2	[+1.2, +9.2]	0.015	0.106	0.016
TrustRAG (filter)	500	+5.8	[+2.0, +9.6]	0.003	0.022	0.004
MADAM-RAG	500	+6.8	[+2.6, +11.0]	0.002	0.020	0.002
CAG (prompted)	500	+7.8	[+4.0, +11.6]	<0.001	0.010	<0.001
ACCU (no copy det.)	500	+6.8	[+2.6, +10.8]	<0.001	0.010	0.001
RAMDocs, Qwen3-8B						
Vanilla RAG	500	+10.4	[+6.4, +14.4]	<0.001	0.002	<0.001
Near-dup removal	500	−1.4	[−5.8, +3.2]	0.573	0.573	0.597
SemDeDup	500	+6.6	[+2.2, +11.0]	0.004	0.011	0.004
Majority vote	500	+10.0	[+6.0, +14.2]	<0.001	0.002	<0.001
Near-dup + vote	500	+9.2	[+5.2, +13.4]	<0.001	0.002	<0.001
SemDeDup + vote	500	+10.0	[+6.0, +14.2]	<0.001	0.002	<0.001
Astute RAG	500	+28.6	[+24.2, +33.0]	<0.001	0.002	<0.001
InstructRAG-ICL	500	+4.0	[+0.0, +8.2]	0.062	0.123	0.072
TrustRAG (filter)	500	+10.8	[+6.8, +14.8]	<0.001	0.002	<0.001
MADAM-RAG	500	+20.2	[+15.6, +24.8]	<0.001	0.002	<0.001
CAG (prompted)	500	+7.0	[+3.2, +10.8]	<0.001	0.002	<0.001
ACCU (no copy det.)	500	+16.2	[+11.8, +20.8]	<0.001	0.002	<0.001

Table 9: Validity of the string-match metric: an LLM judge (Qwen3-8B) re-grades the same random instances for seven methods (Qwen2.5-7B reader). Agreement and disagreement rates in %; Kendall τ between the method rankings under judge and metric.

Data	n	agree	κ	J+/M−	J−/M+	τ
EchoBench ($m \geq 1$, strict)	700	97.4	0.94	0.9	1.7	0.80
RAMDocs (all valid)	420	81.4	0.56	14.8	3.8	0.92

Table 10: Accuracy (%) with m=4 copies of a single type vs. k=2 independent correct pages.

Method	verb.	b.-tr.	para.	farm	summ.	mixed
Qwen2.5-7B						
Vanilla RAG	70.9	66.3	57.6	63.6	64.0	61.5
Near-dup removal	81.9	69.1	57.6	63.6	64.7	64.0
SemDeDup	79.9	**82.3**	72.3	66.0	66.7	68.0
Majority vote	25.1	26.3	25.0	29.0	32.7	25.5
Near-dup + vote	76.4	30.9	25.0	29.6	34.0	32.0
SemDeDup + vote	75.4	65.7	53.8	33.3	59.3	40.5
TrustRAG (filter)	72.4	68.0	58.2	63.6	64.0	62.0
ACCU (no copy det.)	25.1	26.3	25.0	29.0	32.7	25.5
IndieVote-H	76.9	78.3	77.7	73.5	75.3	76.0
IndieVote-L	76.9	77.7	77.7	75.9	77.3	77.0
IndieBayes-L	76.9	78.3	77.7	75.3	77.3	76.0
IndieRAG-L	**82.4**	79.4	**80.4**	**84.0**	**85.3**	**79.5**
IndieVote-oracle	76.9	78.3	77.7	75.9	78.0	77.0
Qwen3-8B						
Vanilla RAG	76.9	77.1	64.7	66.7	67.3	70.0
Near-dup removal	**87.4**	77.7	64.7	66.7	68.0	72.0
SemDeDup	85.9	**85.1**	77.2	67.3	79.3	78.5
Majority vote	28.6	29.7	32.1	31.5	29.3	32.0
Near-dup + vote	81.9	34.3	32.1	32.1	31.3	38.5
SemDeDup + vote	80.4	72.6	59.8	34.6	60.0	46.0
TrustRAG (filter)	78.4	79.4	67.9	66.7	68.7	70.5
ACCU (no copy det.)	28.6	29.7	32.1	31.5	29.3	32.0
IndieVote-H	81.9	**85.1**	82.6	80.9	81.3	**83.5**
IndieVote-L	81.9	**85.1**	83.2	82.1	83.3	83.0
IndieBayes-L	81.9	**85.1**	82.6	82.1	83.3	82.5
IndieRAG-L	82.9	82.9	**85.3**	**84.0**	**84.7**	78.5
IndieVote-oracle	81.9	85.1	83.2	82.7	83.3	83.5

Table 11: Reader family and scale (key methods; Granite and 32B were run on setting A and B only). Clean and echo-avg accuracy (%).

	Qwen2.5-7B		Qwen3-8B		Granite-3.3-8B		Qwen2.5-32B	
Method	clean	echo	clean	echo	clean	echo	clean	echo
Vanilla RAG	**80.7**	61.0	**85.2**	68.5	**85.8**	67.0	88.0	72.1
Majority vote	75.7	35.0	78.2	40.6	77.2	43.1	80.5	51.7
Astute RAG	76.5	67.4	70.2	46.3	35.5	27.1	**94.0**	84.4
IndieVote-L	75.8	63.6	78.0	67.1	77.2	**68.6**	80.5	69.6
IndieBayes-L	75.8	63.1	78.0	66.9	77.2	68.4	80.5	69.6
IndieRAG-L	80.3	**78.3**	84.7	**79.5**	63.2	63.0	90.7	**87.3**

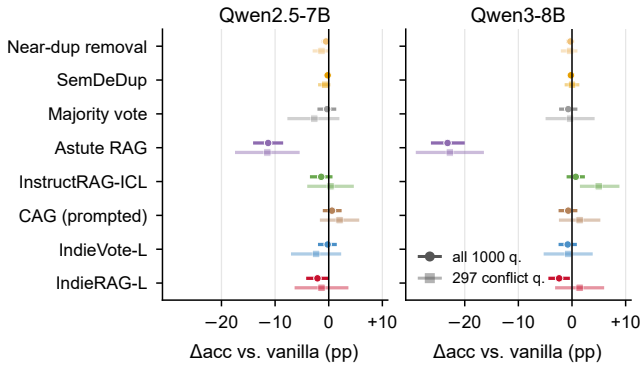


Figure 6: Natural web pages: paired accuracy difference to vanilla RAG with 95% bootstrap CIs, on all 1,000 questions (dark) and the 297 questions whose pages conflict (light).

Table 12: Cost per EchoBench instance (Qwen2.5-7B). Claim extraction (one short call per document) and CAG's per-document credibility ratings are counted as if run standalone. Seq.: sequential LLM rounds; dep.: dependence estimation (ms, CPU); Echo: echo-avg accuracy.

Method	Calls	Prompt	Gen.	Seq.	dep.	Echo
Vanilla RAG	1.0	2.5k	8	1	–	61.0
Near-dup removal	1.0	2.3k	8	1	–	64.0
SemDeDup	1.0	2.2k	8	1	–	66.2
Majority vote	8.5	3.0k	17	1	–	35.0
Near-dup + vote	8.5	3.0k	17	1	0	42.7
SemDeDup + vote	8.5	3.0k	17	1	0	46.8
Astute RAG	2.0	2.6k	189	2	–	67.4
InstructRAG-ICL	1.0	2.5k	169	1	–	66.8
TrustRAG (filter)	1.0	2.5k	8	1	–	62.2
MADAM-RAG	18.9	11.2k	726	3	–	–
CAG (prompted)	8.8	5.2k	26	2	–	68.1
ACCU (no copy det.)	8.5	3.0k	17	1	–	35.0
IndieVote-H	8.5	3.0k	17	1	3	63.1
IndieVote-L	8.5	3.0k	17	1	6	63.6
IndieVote-LLM	19.1	10.0k	27	1	121	59.2
IndieBayes-L	8.5	3.0k	17	1	6	63.1
IndieRAG-H	8.8	4.6k	79	2	3	77.5
IndieRAG-L	9.5	4.9k	81	2	6	78.3
IndieRAG-L+meta	8.8	4.6k	80	2	5	**79.7**
IndieRAG-LLM	20.1	11.9k	91	2	121	77.5

Table 13: Controls and stress tests (Qwen3-8B, accuracy % unless noted). *Style*: correct pages also LLM-rewritten (m=8). *Rank.*: documents in dense-retriever order (m=8). *Noise*: 10 distractor pages (m=4). *B-agr.*: follows the independent majority when wrong sources are independent. *Setting C*: copies of the *true* page vs. w independent wrong pages; C-ind.: agreement with the independent majority at m=4, w=2. Questions per condition: style 183, ranked and noise 200, setting C 93, 98 and 68. Shaded: ours.

	Echoed misinformation				Setting C (echoed truth)		
Method	Style	Rank.	Noise	B-agr.	$m4, w1$	$m2, w2$	C-ind.
Vanilla RAG	59.6	64.5	67.0	59.0	89.2	81.6	10.3
SemDeDup	67.8	–	72.0	59.0	–	–	–
Majority vote	27.9	27.0	34.0	**68.4**	**95.7**	**89.8**	1.5
SemDeDup + vote	34.4	36.5	47.5	**68.4**	**95.7**	82.7	2.9
Astute RAG	18.0	38.5	–	38.9	64.5	55.1	0.0
CAG (prompted)	71.0	–	–	56.6	91.4	82.7	5.9
IndieVote-L	**78.7**	80.0	**83.5**	67.5	**95.7**	44.9	**48.5**
IndieBayes-L	78.1	**81.0**	83.0	67.5	**95.7**	44.9	47.1
IndieRAG-L	77.0	79.5	80.0	62.1	84.9	64.3	29.4

Table 14: RAMDocs (500 questions with ambiguity, misinformation and noise; no synthetic copies). Acc: all valid answers and no misinformation answer; Wrong↓: a misinformation answer is output.

	Qwen2.5-7B		Qwen3-8B	
Method	Acc	Wrong↓	Acc	Wrong↓
Vanilla RAG	26.0	42.4	30.6	42.4
Near-dup removal	**35.6**	22.2	**42.4**	23.2
SemDeDup	29.4	33.6	34.4	34.8
Majority vote	31.6	45.4	31.0	45.0
Astute RAG	27.6	26.6	12.4	**7.0**
InstructRAG-ICL	27.0	23.4	37.0	30.8
TrustRAG (filter)	26.4	39.8	30.2	40.2
MADAM-RAG	25.4	28.0	20.8	19.6
CAG (prompted)	24.4	29.6	34.0	35.4
ACCU (no copy det.)	25.4	22.0	24.8	21.8
IndieVote-L	33.0	46.4	32.4	46.0
IndieBayes-L	22.6	**20.8**	22.2	20.4
IndieRAG-H	32.4	27.8	40.4	32.4
IndieRAG-L	32.2	27.2	41.0	32.4
IndieRAG-LLM	28.2	32.8	37.8	37.8

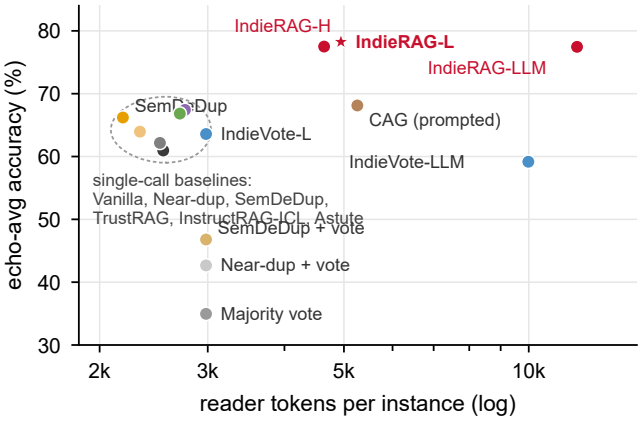


Figure 7: Echo accuracy vs. reader tokens per instance (Qwen2.5-7B; MADAM-RAG omitted as it ran on three conditions).